

# Scramble Inversion as Discrete Denoising Diffusion: A Matched-Compute Study on the Rubik’s Cube Group

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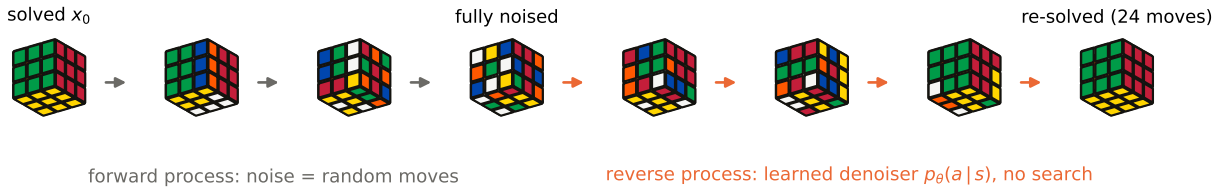


Figure 1: Cube solving as denoising diffusion, end to end (real states, trained model). **Left (gray):** the forward process noises the solved state with random generator moves. **Right (orange):** the learned reverse process—greedy rollout of the trained denoiser, no search—removes the noise move by move; the 24-move solution shown is replay-verified.

## Abstract

Solving the Rubik’s Cube with learned heuristics is conventionally framed as reinforcement learning: deep approximate value iteration (DAVI) learns a cost-to-go function that guides search, as in DeepCubeA. We study an alternative framing in which cube solving is *discrete denoising diffusion on a Cayley graph*: the forward (noising) process scrambles the solved state with random generator moves under a linear depth schedule, and a denoiser is trained by cross-entropy to predict the inverse of the last scramble move—the exact analogue of  $\epsilon$ -prediction. Solving is reverse-process sampling, optionally sharpened by beam search. The training objective coincides with the self-supervised scramble-inversion objective of Takano (2023); our contributions are the diffusion-process formalization, a controlled matched-architecture, matched-hardware comparison against DAVI, and an evaluation protocol with unusually strong guarantees: an exact breadth-first-search oracle over the *entire*  $2 \times 2 \times 2$  group (3,674,160 states) and mechanical replay verification of every claimed solution. On a single consumer GPU, the denoising objective is  $5\times$  cheaper per iteration than DAVI, reaches 99.989% greedy (search-free) solve rate over the full  $2 \times 2 \times 2$  state space where DAVI reaches 77.7%, and solves 1000/1000 fully scrambled  $3 \times 3 \times 3$  cubes after 4.9 hours of training versus 13.9 hours for DAVI at the same solve rate. Exact-oracle diagnostics reveal complementary error structure: the value function degrades monotonically with distance-to-goal, while the denoiser’s per-move accuracy dips precisely where the scramble-length posterior is most ambiguous. Ablations over the noise schedule and timestep conditioning support the diffusion interpretation: a linear (uniform) schedule suffices, and conditioning on the timestep is unnecessary because the group state determines its own noise level. Code and full logs are released for reproducibility.

\*Experiments, implementation, and manuscript preparation were carried out with the assistance of Claude (Anthropic). Correspondence: [aamer.khani.1@gmail.com](mailto:aamer.khani.1@gmail.com).

# 1 Introduction

Learning to solve the Rubik’s Cube is a canonical benchmark for combining neural function approximation with search in enormous discrete state spaces: the  $3\times 3\times 3$  cube group has  $4.3 \times 10^{19}$  elements, a single rewarding state, and no informative local structure that naive exploration can exploit. A forward random walk essentially never finds the solved state, so successful learned solvers manufacture supervision by working *backward* from the goal. DeepCubeA [1] does so with deep approximate value iteration (DAVI): states are sampled by scrambling the solved cube, a cost-to-go network is trained by one-step Bellman backups, and solutions are extracted with value-guided search.

This paper studies a different, arguably simpler framing, proposed independently by the first author from the mechanics of generative diffusion models [3, 9]: treat scrambling as the *forward noising process* of a diffusion model whose data distribution is a point mass on the solved state. Random generator moves substitute for Gaussian noise; the scramble depth is the diffusion timestep; a linear schedule (uniform over depths) replaces the cosine schedule; and the reverse process is a learned denoiser that, given a noised (scrambled) state, predicts the move that undoes the most recent noise increment—the exact discrete analogue of the  $\epsilon$ -prediction objective. Solving a cube is then nothing but reverse-process sampling from the fully-noised regime, optionally sharpened by beam search, which plays the role of a guided sampler.

The resulting training signal—predict the inverse of the last scramble move—is not new: it is precisely the self-supervised objective of Takano [12], which outperformed DeepCubeA on solution optimality at matched search budgets. What is missing from the literature, and what this paper provides, is threefold:

1. **A diffusion-process formalization** (§3) of scramble-inversion training on Cayley graphs, which yields testable predictions: the denoising target is a posterior average over scramble histories (i.e. inherently noisy labels), timestep conditioning should be redundant because a group element determines its own distance-to-goal, and schedule shape should matter only through the marginal state distribution it induces.
2. **A controlled comparison** (§5) against DAVI with matched backbone architecture, batch size, data budget-per-iteration, optimizer, hardware, and evaluation protocol—isolating the training *objective* as the only variable. Prior comparisons (e.g. 12 vs. 1) confound objective, architecture, and compute.
3. **An exact evaluation protocol** (§4): for the  $2\times 2\times 2$  cube we enumerate the entire reachable group (3,674,160 states) by GPU breadth-first search, giving exact optimal distances against which value estimates, per-move decisions, and solution optimality are measured *exhaustively* rather than on samples; and every claimed solution, on either cube, is verified by replaying its move sequence in the environment. To our knowledge no prior learned-solver evaluation reports full-state-space solve rates.

Our findings are summarized as follows. At matched per-sample cost the denoising objective is  $\sim 12\times$  cheaper than a DAVI iteration in forward passes (no  $|A|$ -way neighbor expansion for bootstrapped targets) and  $\sim 5\times$  faster in measured wall clock. On the full  $2\times 2\times 2$  group the denoiser’s search-free greedy rollout solves 99.989% of all states (DAVI: 77.7%), and both methods reach 100% with beam search—the denoiser needing only width 32. On the  $3\times 3\times 3$ , the denoiser solves 1000/1000 hundred-move scrambles after 4.9 h of training (34.9% of them with no search at all), matching DAVI’s solve rate at  $2.8\times$  less wall clock; at every beam width the denoiser

dominates the value function in solve rate per unit of test-time compute (Figure 7). The price is solution length: the denoiser’s likelihood-guided beam yields solutions averaging 29.9 moves against DAVI’s 22.95 in each method’s own best configuration, and the gap closes to  $\sim 0.4$  moves under an identical beam implementation. Exact-oracle diagnostics and ablations (§5.4) corroborate the diffusion view, including a characteristic dip in per-move accuracy at intermediate distances where the posterior over scramble histories is most ambiguous.

## 2 Related Work

**Learned Rubik’s Cube solvers.** DeepCube [6] introduced autodidactic iteration, and DeepCubeA [1] refined it into DAVI: train  $V_\theta(s) \approx d^*(s)$  on reverse scrambles via one-step lookahead targets, then solve with batch-weighted A\*. Non-learned solvers achieve optimality with pattern databases [5]; the cube’s QTM diameter is 26 [8]. Takano [12] trains a policy to predict the scramble-inverting move and decodes with beam search, reporting shorter solutions than DeepCubeA at matched node budgets, with far less training data. Our training objective for the denoiser is the same as Takano’s; we do not claim it as a contribution. Unlike Takano [12], we hold architecture, batch, hardware, and evaluation fixed across both objectives, add exact full-group evaluation, and interpret the method through the diffusion lens with supporting ablations.

**Denoising diffusion.** Diffusion models learn to invert a gradual noising process [3, 9, 10]; discrete-state variants replace Gaussian kernels with categorical transition matrices [2, 4]. Noise schedules are a central design choice in the continuous setting [7]. Diffusion-style iterative refinement has been applied to combinatorial optimization, e.g. DIFUSCO [11], which denoises solution *encodings* for TSP/MIS. Our setting differs structurally: the “noise” is exogenous group action on a Cayley graph, the data distribution is a point mass (the solved state), and the reverse process outputs *actions* rather than solution encodings, making the sampler itself the solver.

## 3 Method

### 3.1 Setup: the cube as a Cayley graph

Let  $G$  be the cube group with identity  $e$  (the solved state) and let  $A = \{a_1, \dots, a_M\}$  be a symmetric set of generators (quarter turns;  $a \in A \Rightarrow a^{-1} \in A$ ). States are represented as sticker colorings  $s \in \{0, \dots, 5\}^S$ ; each generator acts as a fixed permutation of the  $S$  sticker indices, so a batch of moves is a single **gather**. We use the quarter-turn metric throughout. For the  $3 \times 3 \times 3$ ,  $M=12$ ,  $S=54$ . For the  $2 \times 2 \times 2$  we restrict to  $A = \{U^{\pm 1}, R^{\pm 1}, F^{\pm 1}\}$  ( $M=6$ ,  $S=24$ ), which fixes the DLB corner; the reachable set then has exactly  $7! \cdot 3^6 = 3,674,160$  elements with QTM diameter 14, small enough to enumerate exactly (§4). Let  $d^*(s)$  denote the exact distance to  $e$  in the Cayley graph  $\Gamma(G, A)$ .

### 3.2 Forward process: scrambling as noise

The forward process starts at the data distribution—a point mass  $q_0 = \delta_e$  (“in the beginning, the cube is fully built”)—and applies  $t$  random generators:

$$s_0 = e, \quad a_k \sim \text{Unif}(A \setminus \{a_{k-1}^{-1}\}), \quad s_k = a_k \cdot s_{k-1}, \quad k = 1, \dots, t, \quad (1)$$

where excluding the immediate inverse removes the trivial 2-cycle (the discrete analogue of using non-degenerate noise increments). The timestep is drawn from a *linear schedule*  $t \sim \text{Unif}\{1, \dots, K\}$

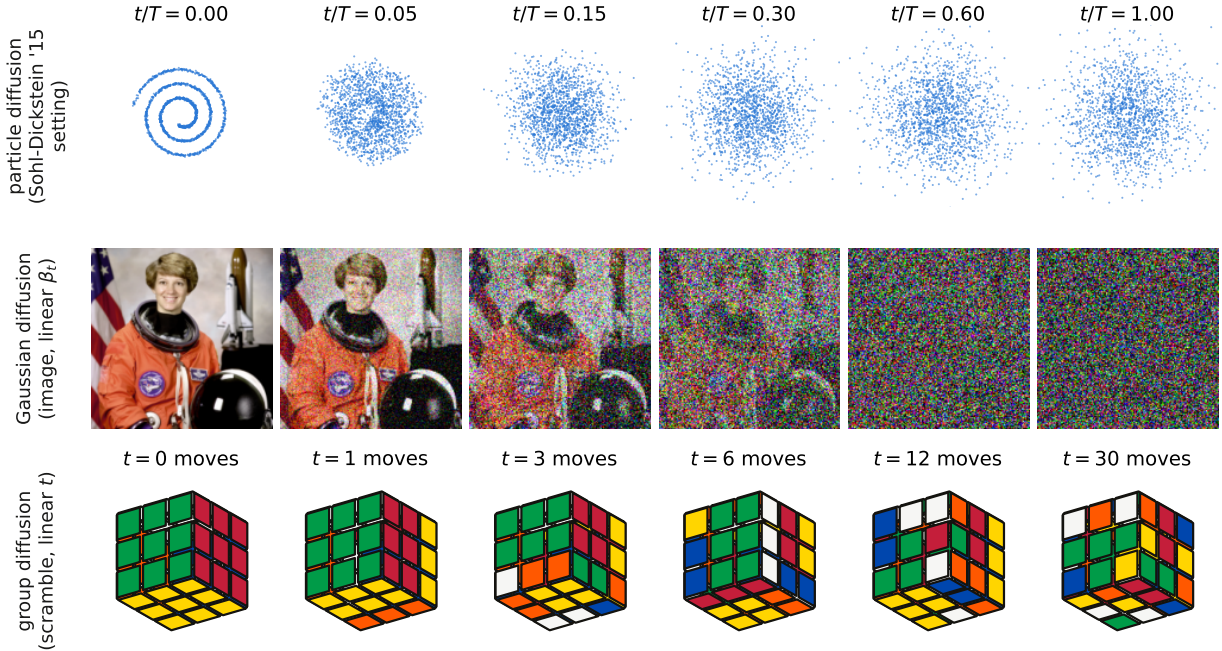


Figure 2: One forward process, three state spaces, at matched noise levels (left to right). **Top:** particle diffusion of a 2-D spiral density into an isotropic Gaussian—the setting in which diffusion models were introduced [9] (our rendering). **Middle:** Gaussian image diffusion under the linear DDPM schedule [3], applied to a standard public-domain test photograph (pixel-space noising, unrelated to the cube). **Bottom:** the group forward process of this paper—the identical cube after  $t$  uniformly random generator moves (actual engine states). In all three, a linear schedule interpolates between the data distribution and the fully-noised regime; on the cube the “noise” is exogenous group action and the fully-noised regime is a uniformly random state.

with horizon  $K$  ( $K=18$  for the  $2 \times 2 \times 2$ ,  $K=30$  for the  $3 \times 3 \times 3$ ; both exceed the respective diameters). Unlike Gaussian diffusion there is no signal-to-noise ratio to shape, so the cosine schedule of Nichol and Dhariwal [7] has no analogue; the schedule matters only through the marginal state distribution  $q_t$  it induces, a claim we test in §5.4. As  $t$  grows past the graph’s mixing scale,  $q_t$  approaches the uniform distribution on  $G$ —the fully-noised regime. Figure 2 places this forward process in the diffusion lineage visually: the same schedule semantics carry from the particle-density setting of the original diffusion formulation through image DDPMs to the cube group, with “pure noise” meaning, respectively, an isotropic Gaussian, a noise image, and a uniformly random group element.

### 3.3 Reverse process: noise prediction and sampling

The denoiser  $p_\theta(a \mid s)$  is trained to predict the inverse of the last applied generator:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, a_{1:t}} \left[ -\log p_\theta(a_t^{-1} \mid s_t) \right], \quad (2)$$

the categorical counterpart of  $\epsilon$ -prediction: the network observes a noised sample and regresses the most recent noise increment. The Bayes-optimal solution of Eq. (2) is the posterior over last-move inverses given the state,

$$p^*(a \mid s) \propto \sum_{t=1}^K \Pr[s_t = s, a_t = a^{-1}], \quad (3)$$

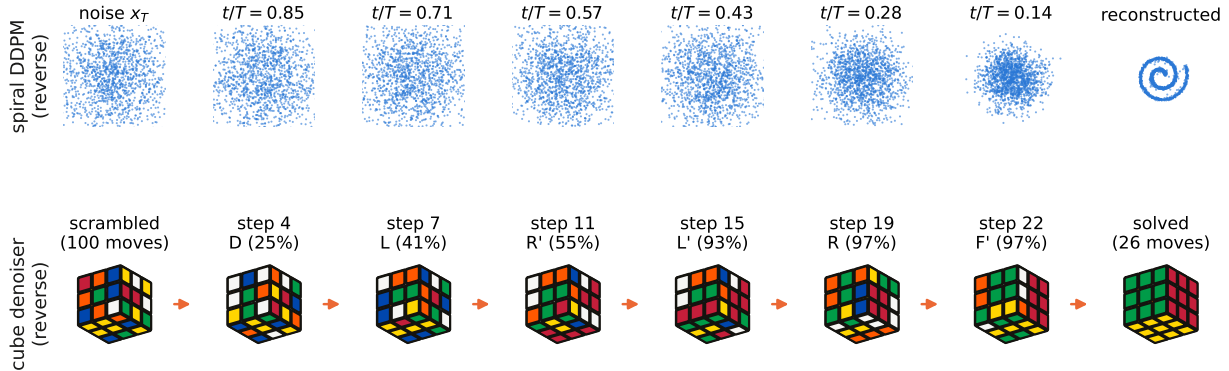


Figure 3: Reverse processes reconstruct the data from noise. **Top:** a small DDPM trained here on the spiral density (the classic reconstruction demonstration of the original diffusion paper [9]) walks 1,600 particles from  $\mathcal{N}(0, I)$  back onto the spiral. **Bottom:** the trained cube denoiser’s *actual* greedy rollout (no search) on a 100-move scramble: each panel shows the chosen move and its probability under  $p_\theta$ ; the 26-move solution is replay-verified. Confidence rises as noise is removed, in both domains.

a *posterior average over scramble histories*. Labels are therefore inherently noisy—a walk that has already wandered back toward  $e$  produces a label pointing away from the goal—but the average correlates strongly with descent on  $d^*$ , and we quantify exactly how strongly in §5 using the oracle. Two properties distinguish this from continuous diffusion and motivate our design:

**No timestep conditioning.** In Gaussian diffusion the denoiser needs  $t$  because a given  $x$  can occur at any noise level. On a Cayley graph the state determines its own noise level:  $d^*(s)$  is a function of  $s$ , and the set of distance-reducing generators  $\arg \min_a d^*(a \cdot s)$  does not depend on how many steps produced  $s$ . We therefore drop  $t$ -conditioning and validate this choice by ablation (§5.4).

**No closed-form marginal, biased shell coverage.** Gaussian diffusion reaches any noise level in one jump with the correct marginal by construction. Here  $q_t$  must be realized by walking, which is cheap ( $t \leq K$  gathers) but *biased*: random walks backtrack, so deep distance shells are reached far less often than their share of the group. Both objectives inherit this bias; we observe its signature in the error structure of both methods (Figure 5).

**Samplers.** Solving is reverse-process sampling from an arbitrarily noised state. We use (i) *greedy ancestral* decoding: iteratively apply  $\arg \max_a p_\theta(a | s)$  with the immediate backtrack masked, and (ii) *beam search* of width  $W$  maximizing the cumulative log-likelihood  $\sum_i \log p_\theta(a_i | s_i)$ —approximate MAP decoding of the reverse trajectory, the discrete counterpart of a guided sampler. Search is an optional sharpener, not a crutch: we report all results at  $W=1$  as well. Figure 3 shows both reverse processes side by side—a small DDPM we train on the classic spiral density reconstructing it from Gaussian noise, and our trained denoiser’s actual greedy rollout reconstructing the solved cube from a 100-move scramble. The denoiser’s confidence rises monotonically as the state approaches the data manifold (25%  $\rightarrow$  97% along the trajectory shown), the discrete analogue of the sharpening posterior in the final steps of Gaussian reverse diffusion.

### 3.4 Baseline: deep approximate value iteration

DAVI [1] trains  $V_\theta(s) \approx d^*(s)$  on the same scramble distribution with bootstrapped one-step targets  $y(s) = 1 + \min_a V_{\bar{\theta}}(a \cdot s)$  (zero at  $e$ ), where the target network  $\bar{\theta}$  is synced when the loss



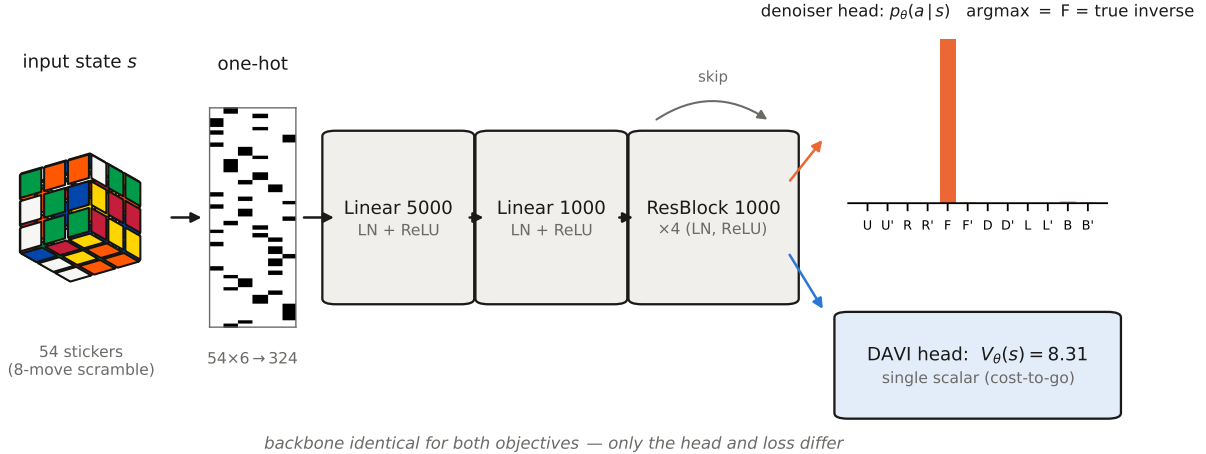


Figure 4: The shared architecture, shown on a *real* forward pass through both trained  $3 \times 3 \times 3$  networks. A state eight scramble moves from solved (left, actual engine state) is one-hot encoded and passed through the common residual-MLP backbone. The trained denoiser head (top right, actual output distribution) places nearly all probability on the true inverse of the last scramble move; the trained DAVI head (bottom right) reads  $V_\theta(s) = 8.31$  for this 8-move scramble. The two objectives share every parameter except the final linear layer.

falls below a threshold. Each iteration therefore costs  $1+M$  forward passes per sample ( $M$ -way neighbor expansion for the target plus the training pass) versus one for the denoiser. At solve time the value function guides beam search by smallest predicted cost-to-go. Backbone, batch size, optimizer, learning-rate schedule, scramble distribution, and hardware are identical to the denoiser’s (Appendix A); the two methods differ only in output head (1 scalar vs.  $M$  logits) and objective. Figure 4 shows the shared architecture executing a real forward pass through both trained heads.

## 4 Experimental Setup

**Vectorized environment.** All experiments run on one consumer laptop GPU (NVIDIA RTX 5080 Laptop, 16.6 GB). States are int8 tensors  $[B, S]$ ; each generator is a precomputed permutation; stepping  $B$  cubes is one **gather** (measured 0.25–0.5 billion moves/s at  $B=10^5$ – $10^6$  even while training occupies the GPU). Move tables are generated geometrically (rotating sticker positions and normals, then re-matching), not hand-coded.

**Engine and oracle validation.** The engine passes group-theoretic identities (generator order 4, inverse consistency, sexy-move order 6,  $(RU)$  order 105 on the  $3 \times 3 \times 3$ , chirality checks). For the  $2 \times 2 \times 2$ , GPU breadth-first search from  $e$  enumerates the reachable set in 0.6s, reproducing the known group order 3,674,160, QTM diameter 14, and the published distance distribution (e.g. exactly 276 states at depth 14), which pins the move tables exactly and yields an oracle table of  $d^*$  for every state.

**Verification protocol.** A cube counts as solved only if replaying the emitted move sequence from the scramble state in the environment lands exactly on  $e$ . Every number below passed this check; no solver output is trusted.

|                                         | DAVI (value)  | Denoising (ours)     |
|-----------------------------------------|---------------|----------------------|
| Parameters                              | 1.7M          | 1.7M                 |
| Training iterations                     | 40,000        | 40,000               |
| Training wall clock                     | 26 min        | <b>6.5 min</b>       |
| Greedy solve rate, all 3,674,160 states | 77.67%        | <b>99.989 %</b>      |
| avg length (vs. exact optimal)          | 15.05 (10.52) | <b>11.59</b> (10.67) |
| Beam-search solve rate (all states)     | 100.000%      | 100.000%             |
| max beam width needed                   | 32,768        | <b>32</b>            |
| Full-sweep evaluation time              | ~35 min       | ~20 s                |

Table 1: Exhaustive evaluation over the entire  $2 \times 2 \times 2$  group (URF generators). Every solution replay-verified. DAVI’s value-guided and the denoiser’s likelihood-guided beams use their respective native scoring.

**Protocol.** Both methods use residual MLP backbones with LayerNorm (Appendix A): 1.7M parameters for the  $2 \times 2 \times 2$ , 14.7M for the  $3 \times 3 \times 3$ , batch 10,000, Adam,  $10 \times$  LR decay at 80% of iterations, bf16 autocast, `torch.compile` for the  $3 \times 3 \times 3$ . DAVI trains for its standard budget (40k iterations on the  $2 \times 2 \times 2$ ; 250k on the  $3 \times 3 \times 3$ ); the denoiser trains for 40k and 500k respectively, and we additionally report the denoiser at *matched iterations* (250k) and throughout *matched wall clock*.  $3 \times 3 \times 3$  test sets are 1000 fresh cubes scrambled with 100 random moves (far beyond the QTM diameter 26); results are single-seed (see Limitations).

## 5 Results

### 5.1 The full $2 \times 2 \times 2$ group, exhaustively

Table 1 reports both solvers evaluated on *all* 3,674,160 states. The denoiser trains in one quarter of the wall clock, solves 99.989% of the entire group with greedy rollout alone (DAVI: 77.7%), and closes to 100% with a width-32 beam; DAVI requires escalating widths up to 32,768 to solve its final holdout state. Greedy solution lengths are near-optimal for the denoiser (11.59 vs. the exact optimum 10.67 averaged over its greedy-solved set).

**Exact error structure.** Figure 5 uses the oracle to dissect both models. DAVI’s value error grows monotonically with  $d^*$  (MAE 4.1 moves at  $d^*=14$ ): bootstrapped values propagate outward from  $e$  and the walk-induced training distribution under-covers deep shells. The denoiser’s per-move accuracy—the probability that its argmax move strictly reduces  $d^*$ —is instead non-monotone: perfect near the goal *and* at the diameter, with a dip to 70–75% at  $d^* \in \{10, 11\}$ . This is the signature predicted by the posterior-averaging view (Eq. 3): at intermediate distances the posterior over scramble histories passing through  $s$  is maximally ambiguous, so labels are noisiest; at the diameter, by contrast, almost every neighbor of a depth-14 state has depth 13, so nearly any move is correct.

### 5.2 $3 \times 3 \times 3$ : matched compute, full scrambles

Table 2 gives the headline comparison; Figure 6 shows greedy capability versus training wall clock. Both methods ultimately solve 1000/1000 fully scrambled cubes with width-2048 beams. The differences are (i) *training cost*—the denoiser reaches 100% after 4.9 h (and already matches DAVI’s final

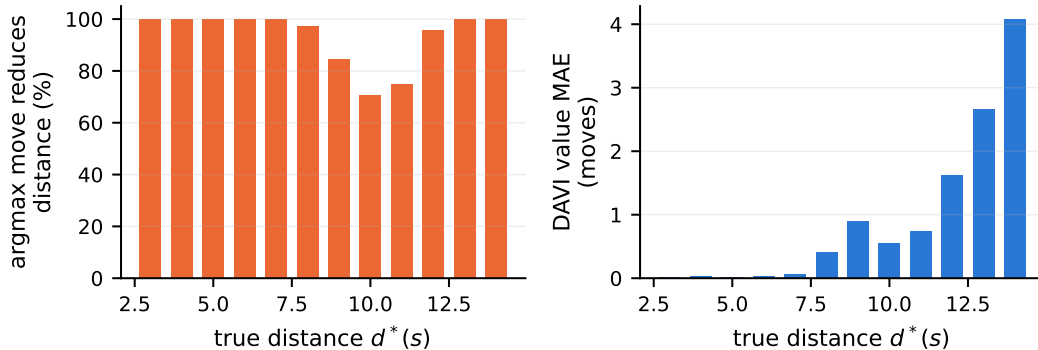


Figure 5: Exact-oracle error structure on the  $2 \times 2 \times 2$  (100k uniform states). **Left:** probability that the denoiser’s argmax move strictly reduces true distance, by  $d^*$ . **Right:** DAVI value MAE by  $d^*$ . The two objectives fail in complementary, mechanistically interpretable ways.

|                                    | DAVI (value) | Denoising (ours)     |              |
|------------------------------------|--------------|----------------------|--------------|
|                                    | 250k iters   | 250k iters (matched) | 500k iters   |
| Parameters                         | 14.7M        | 14.7M                | 14.7M        |
| Wall clock                         | 13.9 h       | <b>2.4 h</b>         | 4.9 h        |
| Iterations/s (steady)              | 5.8          | <b>28.7</b>          | 28.7         |
| Solve rate (1000 cubes, beam 2048) | 100%         | 100%                 | 100%         |
| Greedy-only solve rate             | 0%           | 25.7%                | <b>34.9%</b> |
| Avg solution length                | <b>22.95</b> | 29.02                | 29.88        |
| Max solution length                | <b>28</b>    | 60                   | 60           |
| Eval wall clock (1000 cubes)       | 12.0 min     | <b>~1 min</b>        |              |

Table 2:  $3 \times 3 \times 3$  on 1000 hundred-move scrambles, all solutions replay-verified. Each method uses its native beam scoring (value vs. cumulative log-likelihood); see Figure 7 for the implementation-matched comparison.

solve rate at matched iterations, reached in 2.4h) versus 13.9h for DAVI; (ii) *search dependence*—the denoiser solves 349/1000 cubes with no search whatsoever, while DAVI’s greedy descent solves none (its value plateaus leave greedy trajectories orbiting); and (iii) *solution length*—DAVI’s best configuration produces solutions averaging 22.95 moves (max 28, near the QTM diameter 26) versus 29.88 for the denoiser’s likelihood-guided beam. Length is the one axis on which the value function retains an advantage in its own solver configuration; under an identical beam implementation the gap nearly vanishes (below).

### 5.3 Test-time compute scaling

Figure 7 holds the solver implementation fixed—the same batched, deduplication-free beam for both methods—and sweeps the width. The denoiser dominates at every width: at  $W=512$  it solves 100% of 200 scrambles in 4.1s total, where the value net solves 26.5% in 102s. Two subtleties deserve emphasis. First, solution lengths converge at large widths (23.75 vs. 24.1 at  $W=2048$ ): the denoiser’s longer solutions in Table 2 are a property of likelihood-guided decoding, not of the learned representation. Second, DAVI is far more sensitive to solver implementation: with a sequential beam that maintains a visited-set (deduplicating revisits), DAVI recovers 100% at



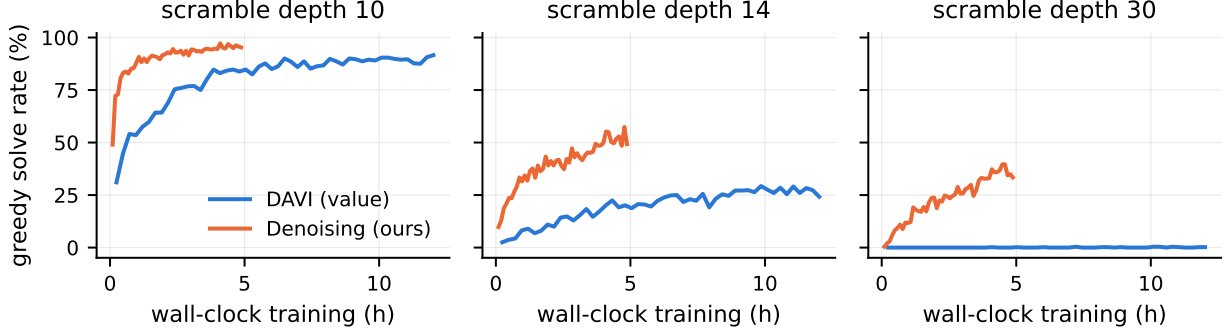


Figure 6: Greedy (search-free) solve rate versus training wall clock on the  $3\times 3\times 3$ , by scramble depth (512 cubes per probe). The denoising objective converts GPU-hours into search-free capability several times faster than value iteration, and is the only method that lifts off at depth 30 within either budget.

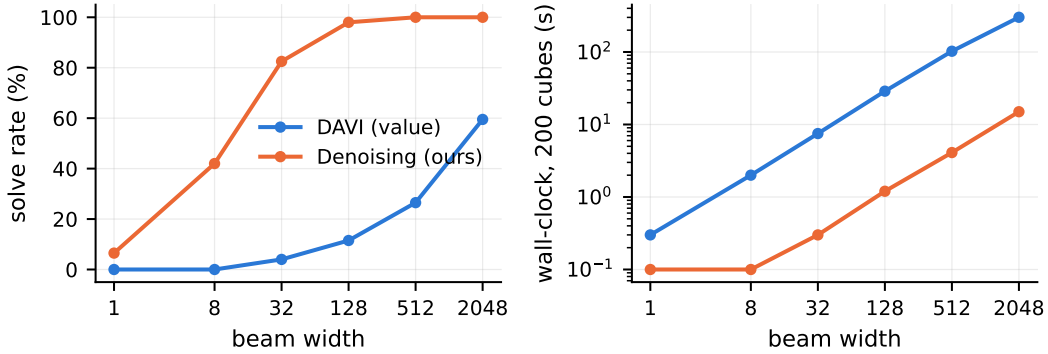


Figure 7: Test-time compute scaling on the  $3\times 3\times 3$  (200 hundred-move scrambles; identical batched no-dedup beam for both methods). **Left:** solve rate vs. beam width. **Right:** wall clock (log scale). The denoiser dominates at every width and is  $\sim 20\times$  faster at equal width because it scores candidates with the parent’s forward pass rather than evaluating every child.

$W=2048$  (Table 2), while the no-dedup beam above reaches only 59.5%. Value-guided beams collapse onto plateaus of near-equal predicted cost unless revisits are pruned; likelihood-guided beams diversify intrinsically because path scores accumulate. We report both configurations for fairness.

#### 5.4 Ablations: schedule and timestep conditioning

Table 3 ablates the two choices most closely tied to the diffusion analogy, on the  $2\times 2\times 2$  where the entire group can be swept—so differences of hundredths of a percent are exact counts of failed states, not sampling noise (though all variants are single training runs). **Schedule.** Every schedule reaches  $\geq 99.96\%$  search-free: the objective is robust to schedule shape, as expected when the schedule matters only through the induced state marginal. The ordering of the residuals is nonetheless systematic and interpretable: shallow-biased sampling leaves  $3.5\times$  more unsolved states than the linear schedule ( $\approx 1389$  vs. 402), deep-biased sampling roughly halves them ( $\approx 239$ ), and per-move accuracy moves far less than coverage does. Deep-shell coverage, not label quality, is the binding

| Variant                  | Schedule          | Greedy solve rate (all states) | $\approx$ unsolved | Move quality |
|--------------------------|-------------------|--------------------------------|--------------------|--------------|
| Linear, $K=18$ (main)    | uniform 1..18     | 99.9891%                       | 402                | 80.7%        |
| Linear, $K=14$           | uniform 1..14     | 99.9865%                       | 496                | <b>82.4%</b> |
| Shallow-biased, $K=18$   | $d \propto$ small | 99.9622%                       | 1389               | 79.3%        |
| Deep-biased, $K=18$      | $d \propto$ large | 99.9935%                       | 239                | 81.6%        |
| $t$ -conditioned, $K=18$ | uniform 1..18     | <b>99.9971%</b>                | 107                | 77.1%        |

Table 3: Noise-schedule and conditioning ablations on the  $2 \times 2 \times 2$ : greedy (search-free) solve rate over all 3,674,160 states (with implied unsolved-state counts) and fraction of argmax moves that strictly reduce exact distance (100k uniform states;  $t$ -conditioned variant queried at  $t=K$ ). 40k iterations each, single run per variant. Shallow/deep bias:  $d = \lceil Ku^2 \rceil$  /  $d = \lceil K\sqrt{u} \rceil$ ,  $u \sim \text{Unif}(0, 1)$ .

constraint. Setting the horizon to the exact diameter ( $K=14$ ) is marginally worse than over-noising ( $K=18$ ) in coverage; as in continuous diffusion, over-noising is cheap insurance (walk depth over-estimates true distance, so  $K$  beyond the diameter costs little). **Timestep conditioning.** The  $t$ -conditioned variant is *not* better calibrated per move—its single-shot move accuracy is the worst of all variants when queried at the fully-noised setting (77.1%)—yet its annealed greedy rollout leaves the fewest unsolved states ( $\approx 107$ ). We read this as  $t$ -annealing acting as a mild decode-time diversity mechanism (the policy is not a fixed function along the rollout, so deterministic cycles are broken), not as evidence that the denoiser needs the noise level: dropping  $t$  entirely still solves 99.989% with a strictly simpler model. The state carries its own noise level; conditioning on  $t$  is optional, not necessary.

### 5.5 Why does the denoiser need so much less search?

DAVI must *rank* neighbors by scalar value; adjacent values differ by at most 1, so ranking errors appear as soon as value MAE approaches 0.5, which happens by  $d^* \approx 8$  (Figure 5, right). Greedy descent then cycles on plateaus, and beams without deduplication drown in value ties. The denoiser sidesteps ranking through a direct *classification* of the descent direction: even with 19% noisy labels its argmax move makes progress at nearly every distance, so unguided rollouts mostly reach the goal, and shallow beams absorb the residual label noise. In diffusion terms: learning the score of the reverse kernel is easier than learning the potential whose gradient it is.

## 6 Discussion and Limitations

**What the analogy buys.** The diffusion framing is not decoration: it predicted (i) that timestep conditioning is unnecessary, (ii) insensitivity to schedule shape beyond induced coverage, (iii) the noisy-label structure of the objective and where it should bite (the mid-distance dip), and (iv) that over-noising ( $K$  beyond the diameter) is safe. All four held empirically, with one instructive nuance on (i): while the unconditioned denoiser loses nothing in per-move accuracy, *annealing* an optional  $t$  input at decode time breaks deterministic rollout cycles and further shrinks the (already tiny) set of greedy failures—a decode-time effect, not a representation requirement (§5.4).

**What it does not buy.** There is no closed-form marginal: deep shells are exponentially under-sampled by walks, a bias shared with DAVI and visible in both error profiles. Likelihood-guided decoding optimizes plausibility, not brevity—solution length is the one metric where the value function’s own solver wins (22.95 vs. 29.88); combining both signals (denoiser proposals scored by a value function) is an obvious hybrid we leave to future work.

**Scope.** Results are single-seed on one hardware platform; run-to-run variance is not characterized, although the exhaustive  $2 \times 2 \times 2$  sweep eliminates test-set noise entirely on that puzzle. We evaluate in QTM only, on cubes only, and at a scale (14.7M parameters,  $\leq 5 \times 10^9$  samples) far below DeepCubeA’s original training; absolute solution-length comparisons to published DeepCubeA numbers are therefore not meaningful. The  $2 \times 2 \times 2$  uses a restricted generator set (URF), standard but different from the full-move convention. Solution lengths for the denoiser are bounded by our rollout cap (60 moves).

## 7 Reproducibility

All code is a small self-contained repository ( $\sim 1500$  lines: vectorized environment with geometric move-table generation and group-identity tests, GPU BFS oracle, both trainers, batched solvers with replay verification, and every evaluation and figure script), released at <https://github.com/aamirkhani/rubiks-diffusion>. All training logs and evaluation JSON reports referenced here are included in the repository and are regenerated by the released scripts; hyperparameters are listed in Appendix A. Total compute for every experiment in this paper:  $\sim 22$  GPU-hours on one laptop GPU.

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## A Hyperparameters and matched settings

|                               | 2×2×2                                                                                        | 3×3×3                             |
|-------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------|
| Backbone                      | MLP: one-hot $\rightarrow h_1 \rightarrow h_2 \rightarrow$ residual blocks (LayerNorm, ReLU) |                                   |
| $h_1, h_2$ , blocks           | 1024, 512, 2                                                                                 | 5000, 1000, 4                     |
| Output head (DAVI / denoiser) | scalar / 6 logits                                                                            | scalar / 12 logits                |
| Parameters                    | 1.7M                                                                                         | 14.7M                             |
| Batch size                    | 10,000                                                                                       | 10,000                            |
| Optimizer                     | Adam, lr $10^{-3}$ , $\times 0.1$ at 80% of iterations                                       |                                   |
| Precision                     | bf16 autocast (+ <code>torch.compile</code> on 3×3×3)                                        |                                   |
| Scramble horizon $K$          | 18 (DAVI: 14 then 18)                                                                        | 30                                |
| Iterations (DAVI / denoiser)  | 40k / 40k                                                                                    | 250k / 500k                       |
| DAVI target-net sync          | loss $< 0.05$ checked every 100 iters; forced after 2000/5000                                |                                   |
| Greedy rollout cap            | 40 moves                                                                                     | 60 moves                          |
| Beam widths (main eval)       | 32-32,768                                                                                    | 2048                              |
| Test scrambles                | entire group                                                                                 | 1000 cubes, 100 moves, fixed seed |
| GPU                           | NVIDIA RTX 5080 Laptop, 16.6 GB                                                              |                                   |

Table 4: Matched configurations. The two methods differ only in output head and training objective.

## B Training-distribution statistics

Measured over  $10^6$  forward-process samples per puzzle: the scramble walk terminates *at* the solved state in 0.07% (2×2×2) and 0.01% (3×3×3) of samples; at least one neighbor of the sampled state is the solved state in 7.6% and 3.4% respectively. Under the denoising objective every sample carries a defined label regardless; under DAVI these near-goal encounters are the sole source of grounded (non-bootstrapped) value signal. For contrast, a *forward* random walk from a scrambled 3×3×3 state has  $\Pr[\text{reach } e] \approx 2.3 \times 10^{-20}$  per step-sequence sample—the quantity that makes naïve exploration-based RL infeasible on this domain.